

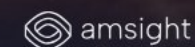
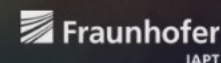
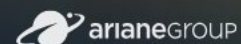
Materials built for the extreme.

Designed with AI. Made and tested in Europe.

FUNDED BY THE EUROPEAN SPACE AGENCY

PRISM's first deployment, through ESA's programme for future launchers (FLPP).

PRISM ALPHA, WITH



Vulcan 2 on test stand P5, DLR Lampoldshausen. Photo: DLR, CC BY 3.0.

THE PROBLEM

Far too many possible alloys to make them all.

474,311,376

possible alloys from five of nine high-melting metals, mixed in 1% steps

130,000 years

to make each one once, at ten a day

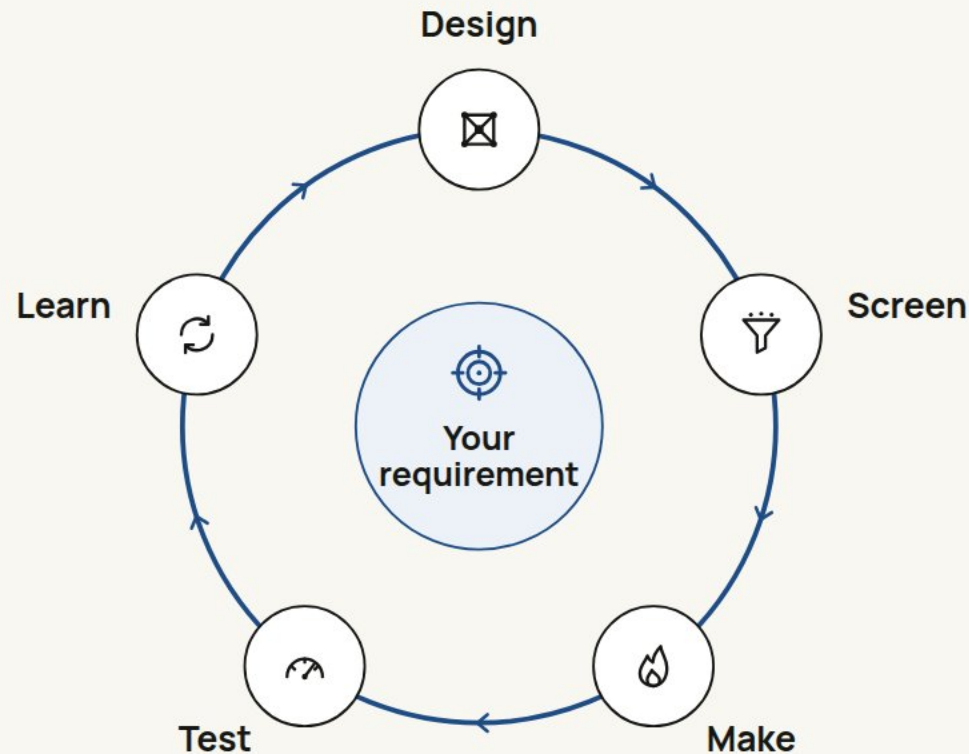
10–20 years

for a new material to reach service today

Sources: Arithmetic: $126 \text{ ways to pick five of nine metals that melt above } 1,650^\circ\text{C} \times 3,764,376 \text{ mixes in whole percent}$. The field is to scale; its layout is illustrative.



One loop: design, make, test, learn.

**Requirement**

You tell us what the part must survive.

PROTOTYPE

**Design**

AI suggests new mixes, from the whole range.

PROTOTYPE

**Screen**

Simulations rule most out, before anything is melted.

PROTOTYPE

**Make**

We melt and 3D-print the best ideas.

IN USE

**Test**

Each sample is measured against your targets.

IN USE

**Learn**

Every result goes back into the models.

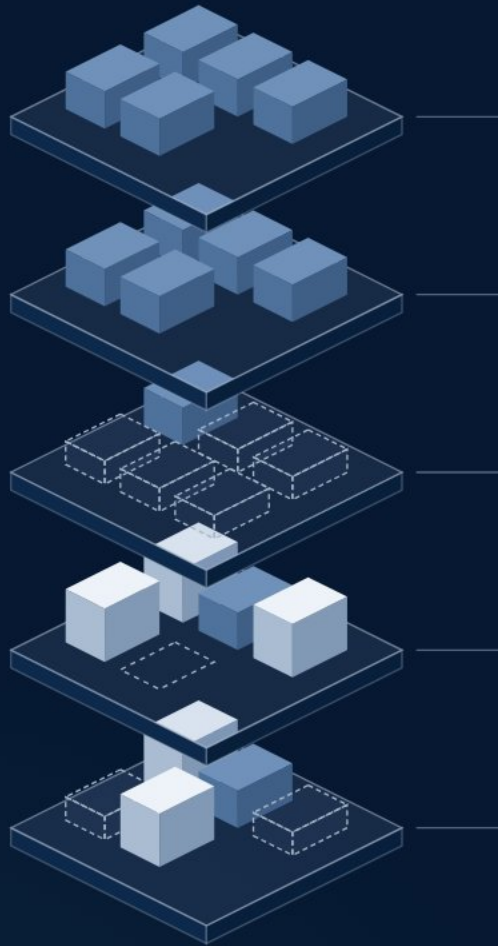
PROTOTYPE

FIRST TARGET

Refractory high-entropy alloys for oxygen-rich preburners in rocket engines, to replace Monel K500 and other legacy alloys.

Five stacks, one system.

■ In use ■ Prototype □ In development --- Goal



01 Research

AI suggests mixes; physics simulations throw out what cannot work.

PROTOTYPE

02 Harness

Plans each round, runs the tools, scores the results, remembers failures.

PROTOTYPE

03 Autonomy

Robots weigh, melt and measure. People stay in charge.

IN DEVELOPMENT

04 Manufacturing and test

Melting, metal 3D printing and testing, with Bimo Tech and Fraunhofer IAPT.

IN USE

05 Evidence

Every result keeps its source, its owner and its rules.

IN USE

Maturity: Each block on a plate is one component, as tall as it is built. In use: runs in current programmes. Prototype: PRISM software, being taken from TRL 3 to 4 in PRISM Alpha. In development: being built.

Four modules, one loop.



Loop: The planner sends a campaign; the samplers propose candidates; simulation and the lab test them; results are kept, with sources; what worked and what failed plans the next round. An engineer signs off what leaves.

Physical synthesis supplies the reward signal the models train on.

01 Recipe translation

PRISM outputs target recipes: precursors, temperatures and thermal profiles.

02 Robotic synthesis

Robot arms dose powder and move samples through heating profiles.

03 Automated characterisation

Diffraction patterns are captured from every synthesised sample.

04 Phase identification

Neural networks compare patterns against structure databases to confirm phases.

05 Feedback

Structured results return to the Evolver's reflector and change the next campaign.

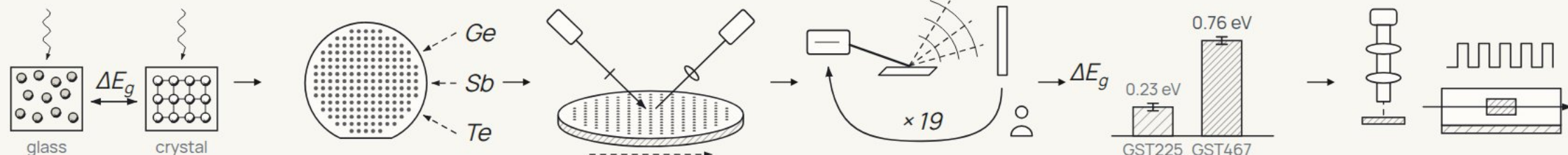
Maturity: In development. Direct instrument control closes the loop in phase III of the plan.



The melt spinner in the materials lab at WUST, Wrocław, where our alloys are melted. We rent the equipment.

Every result carries its own proof.

A REAL LINEAGE, PUBLISHED BY NIST IN 2020



01 · THE QUESTION

Which Ge-Sb-Te mix changes most between glass and crystal?

○ PUBLIC

02 · THE OPTIONS

177 mixes, side by side on one wafer.

○ PUBLIC

03 · WHAT WAS KNOWN

A light scan of every spot, before the run.

○ PUBLIC

04 · THE MEASUREMENTS

19 rounds. Each time, the AI picked what to X-ray.

○ PUBLIC

05 · THE ANSWER

$\text{Ge}_4\text{Sb}_6\text{Te}_7$: about three times the contrast of GST225.

○ PUBLIC

06 · THE PROOF

Confirmed under an electron microscope and in a working device.

○ PUBLIC

CAMEO, 19 rounds

about 10 hours



Measuring all 177

more than 90 hours



RIGHTS TRAVEL WITH EVERY RECORD

Published data ○ PUBLIC

Partner data ■ PRIVATE

Result ■ PRIVATE

A result keeps the strictest rights of its inputs.

FOUR LEVELS OF SHARING

■ PRIVATE

Stays with its owner.

○ RELEASED

Shared with named partners.

☑ COMPUTABLE

Software may use it. No person reads it.

○ PUBLIC

Published on purpose.

WHAT WE PROTECT

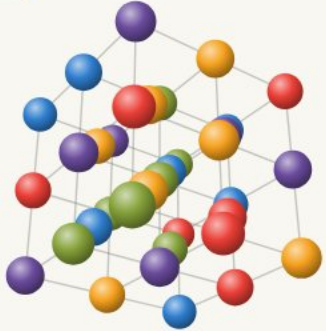
Generative sampling, conditioned on thermodynamic feasibility, reaches metastable alloys that equilibrium screening excludes.

PATENT FILINGS IN PROGRESS

Sources: CAMEO is the work of NIST and its partners, not ours: Kusne et al., Nature Communications 11, 5966 (2020). PRISM keeps the same record for every result: provenance and export labels on ESA work today, automatic enforcement of sharing rules in development.

Funded, contracted and in the lab.

ACTIVE



Project SPARK

ESA activity · Bimo Tech

8

Refractory high-entropy alloy candidates, taken to two physical down-selections.

RUNNING



TECHNOLOGY READINESS LEVEL

PRISM Alpha

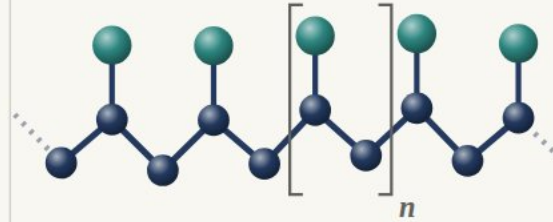
ESA · future launchers programme (FLPP)

TRL 3 → 4

In 12 months: at least three candidates and one complete closed loop.

With  arianespace  Fraunhofer IPT  amsight

CONTRACTED



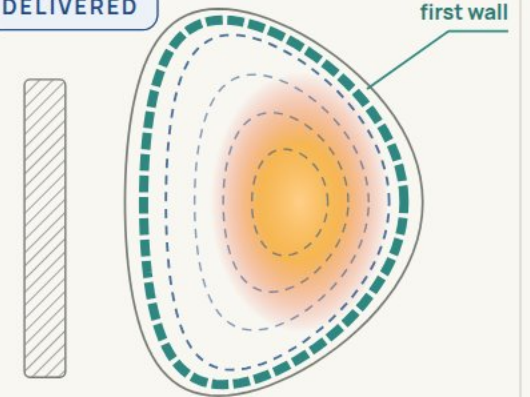
PFAS-free polymers

Industrial partner · under NDA

1st

Privately funded programme, and our first polymer class. Contract and NDA signed; start pending.

DELIVERED



Fusion heritage

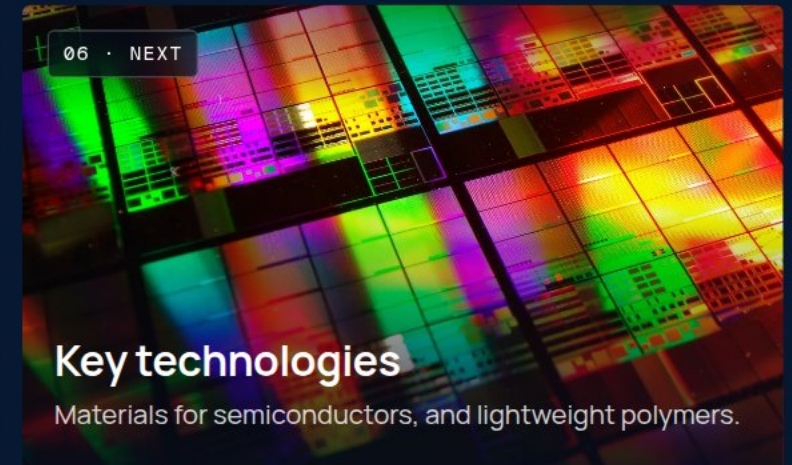
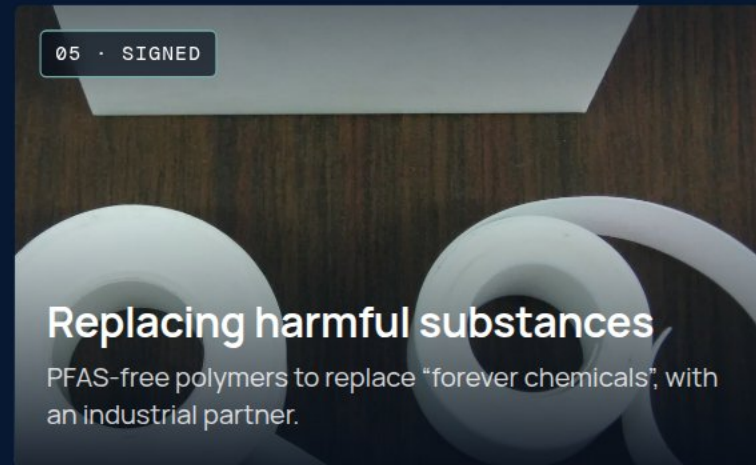
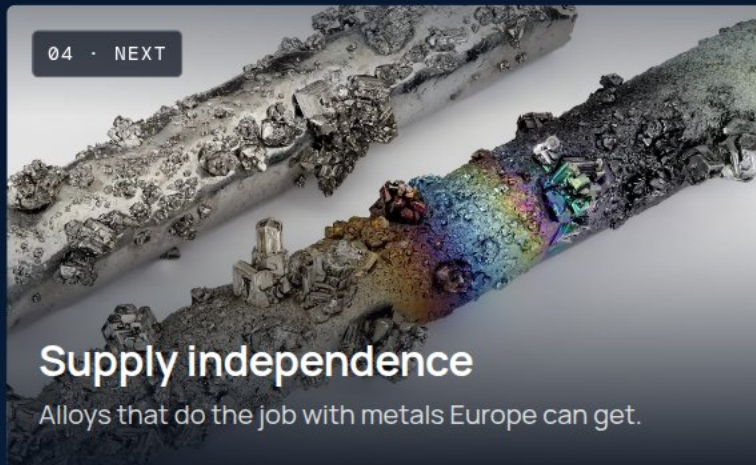
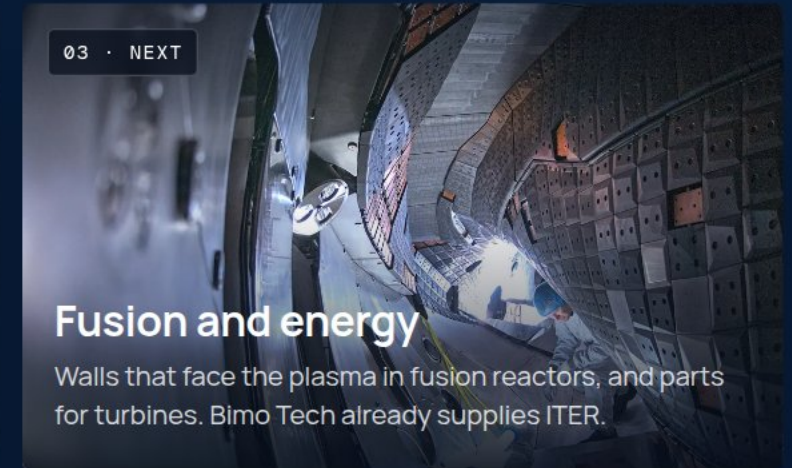
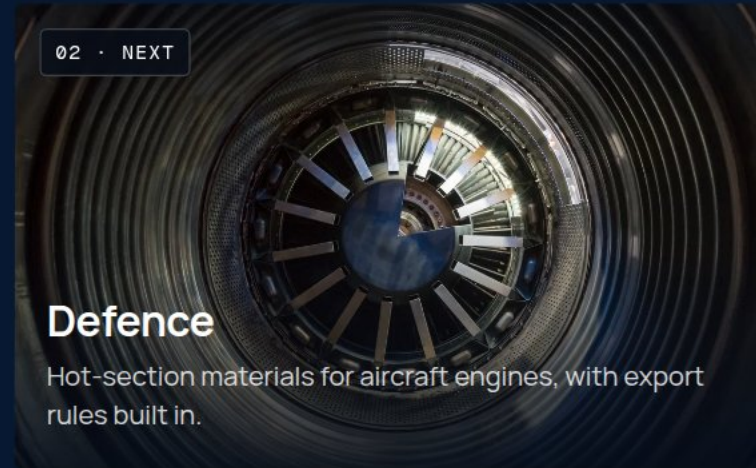
Bimo Tech · ITER

ITER

Titanium first-wall materials and rhodium targets, supplied by Bimo Tech.

Sources: ESA contract records; Mirdyne and Bimo Tech programme records.

Where PRISM goes first.



Photos: Aestus engine in test stand P4.2: DLR, CC BY 3.0 · EJ200 afterburner: Julian Herzog, CC BY 4.0 · Wendelstein 7-X wall tiles: Christopher Roux, EUROfusion, CC BY 4.0 · Tungsten crystals: Alchemist-hp (pse-mendelejew.de), Free Art License. All cropped.

€48M

Serviceable annual wallet

Base case: the upper bound on annual spend across reachable accounts.

€170.7B

EU aerospace
manufacturing
turnover
MKT3

€8.84B

European
space
manufacturing
final sales
MKT4

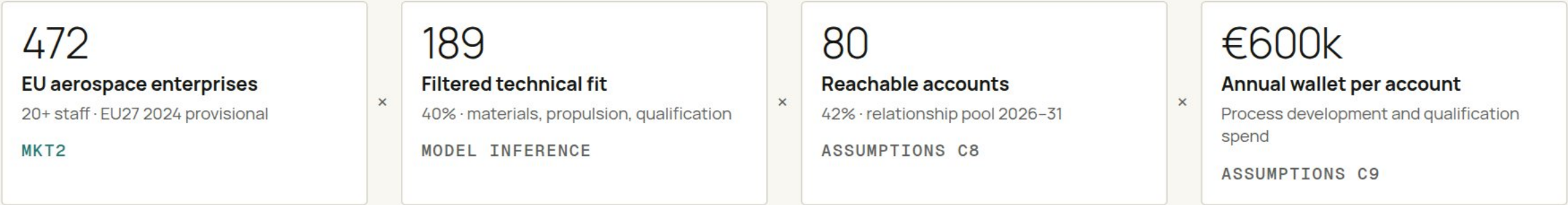
€8.26B

ESA annual
budget, 2026
MKT6

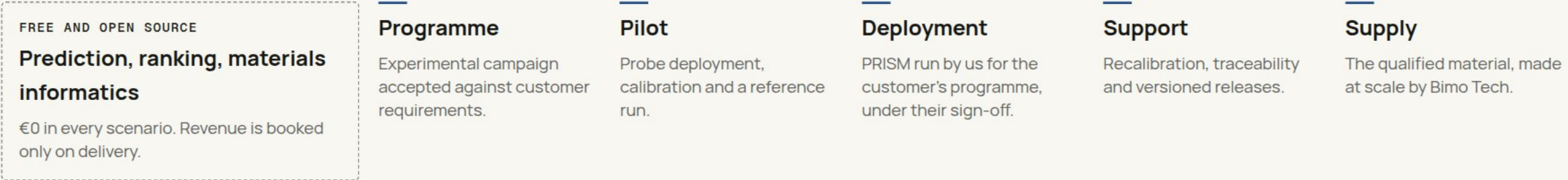
69.6%

Public-
customer
share of space
sales
MKT5

SERVICEABLE ACCOUNT BRIDGE

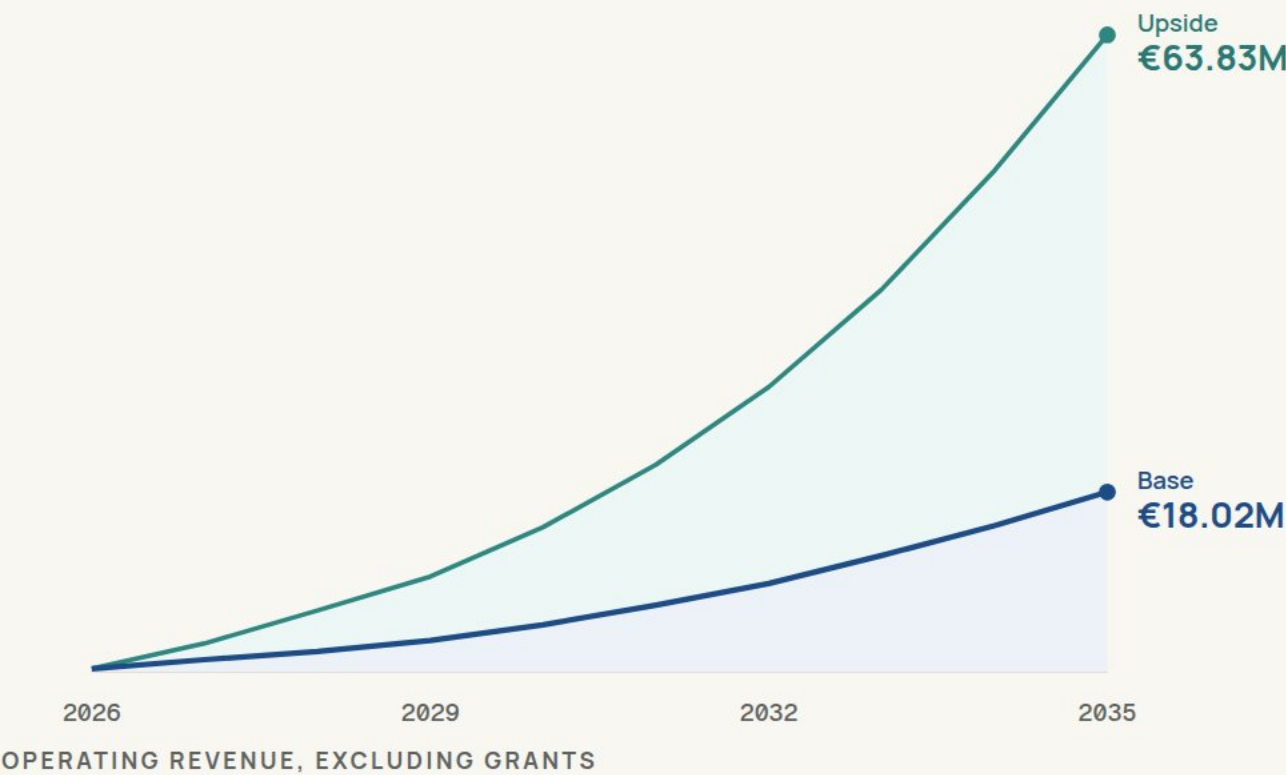


MONETISATION BOUNDARY



Base case: €18.0M revenue by 2035 · EBITDA-positive from 2028 · 61% gross margin. External figures carry their source IDs from the model's evidence register.

Ten-year operating plan.



	BASE	UPSIDE
Operating revenue 2035	€18.02M	€63.83M
EBITDA 2035, before grants	€4.71M	€40.84M
Gross margin 2035	61%	81%
EBITDA-positive from	2028	2027
Cumulative equity required	€2.91M	€0.58M

Revenue is capacity-constrained in both cases: programmes are recognised only up to the delivery capacity left after pilots, deployments and support, so the hiring plan sets the growth rate.

Source: PRISM financial model, research refresh 27 Jul 2026 · checks pass · € nominal.

The team that runs PRISM's ESA work.

KG

Kevin Grüning

MANAGING DIRECTOR, MIRDYNE

Space Systems Lead at Bimo Tech. Physics and technology for space applications, JLU Giessen and THM.

SK

Siddhartha Yash Kovid

TECHNICAL LEAD, MIRDYNE

Technical lead of the ESA projects SPARK and PRISM Alpha at Bimo Tech. Applied AI and data science, MIT Professional Education; biomedical engineering, THM.

MO

Marcin Orzechowski

CO-FOUNDER, MIRDYNE

CEO and Head of R&D at Bimo Tech, which supplies special metals and precision parts for space, energy and science. Wrocław University of Technology.

**KI-Sonderpreis, Hessen Ideen 2026**

The special prize for artificial intelligence. Photo: Hessen Ideen.



Mirdyne is a spin-off of Bimo Tech, which makes special metals and precision parts and supplies ITER.

€4M

Seed round, closing 15 November 2026.

The plan needs €2.91M never to run out of cash. We are raising €4M so the plan survives being wrong, and so the downside gate closes before the money does.

ALLOCATION

Team 65%

Delivery, materials science, product and security headcount

Go-to-market, IP and export control 16%

Bids, consortia, field selling, patents, dual-use compliance

Compute, infrastructure and lab access 13%

Controlled core, per-site secure compute, partner-lab access

Probe hardware, tools and testing 6%

Probe builds, calibration, R&D tooling

ROADMAP · 24 MONTHS

- I **Computational validation**
6 months. The full discovery loop running the first PRISM campaigns.
- II **Hybrid loop**
12 months. Integration with Fraunhofer IAPT additive-manufacturing data.
- III **Autonomous laboratory**
24 months. Direct instrument control closes the loop with no human in the sequence.

Start with the capability you need.

Alloys and polymers, designed, made and tested in one loop, with the proof attached.

Register interest →

mirdyne.com →

The European Space Agency funds PRISM's first deployment, through its programme for future launchers (FLPP).

